

Operator precedence / Operator priority

(ASSOCIATION OF OPERATORS)

1. ()
2. +, -, ! (sign operators, unary operators)
3. ++, -- (pre increment & decrement)
4. *, /, %
5. +, - (Binary)
6. ==, !=
7. &&
8. ||
9. ?: (ternary operator)
10. =
11. ++, -- (Post increment & decrement)
12. , (comma)

/ - division [Quotient]:

$$5/2=2$$

Note: In C & C++ both operands are int then result also int. if anyone or both are floats then result also float.

$$5.0/2=2.500000$$

$$5/2.0=2.500000$$

$$5.0/2.0=2.500000$$

$$(\text{float})5/2=2.500000 \text{ [explicit type casting]}$$

$$(\text{int}) 5.0/2=2 \text{ [explicit type casting]}$$

$$\text{int } a=3.5; \rightarrow a=3 \text{ [implicit type casting]}$$

$$\text{float } b = 12; \rightarrow b=12.000000 \text{ [implicit type casting]}$$

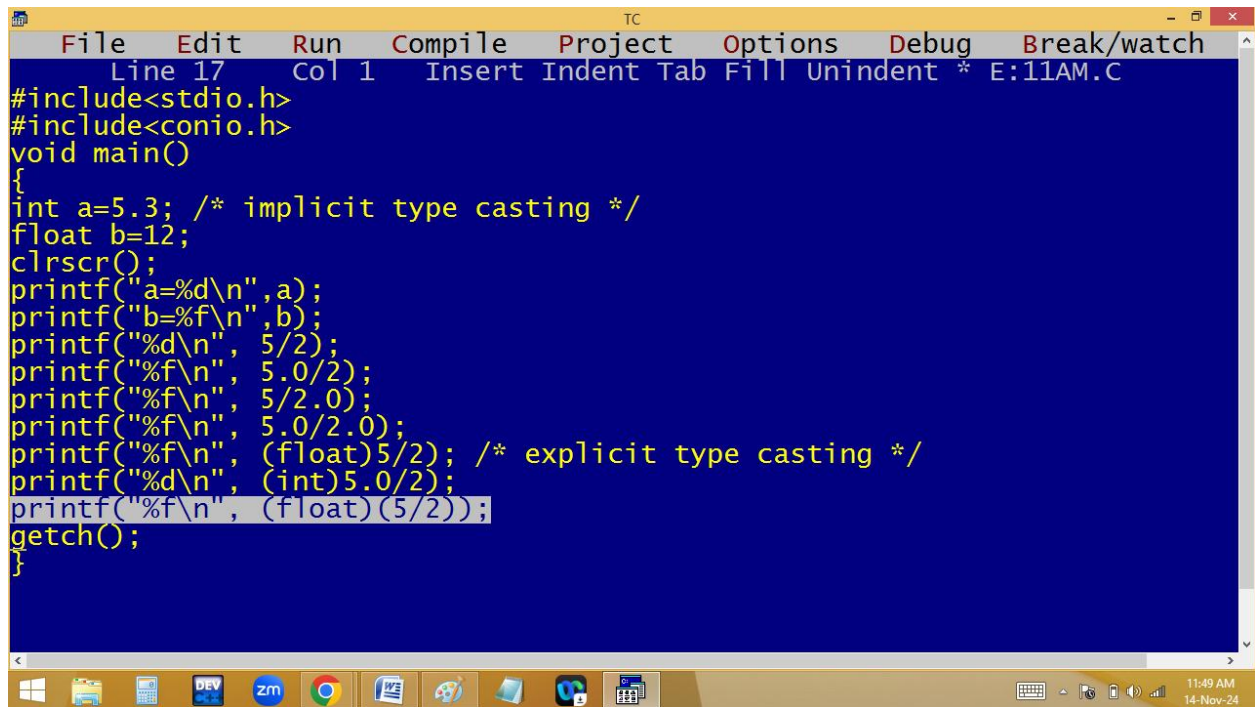
$$\text{float } x = 5/2; \rightarrow x= 2.000000$$

```
≡ File Edit Search Run Compile Debug Project Options Window Help
[ ] NONAME00.CPP 1=[↑]
#include<stdio.h>
#include<conio.h>
void main()
{
float x=5/2;
clrscr();
printf("x = %f",x);
getch();
}_

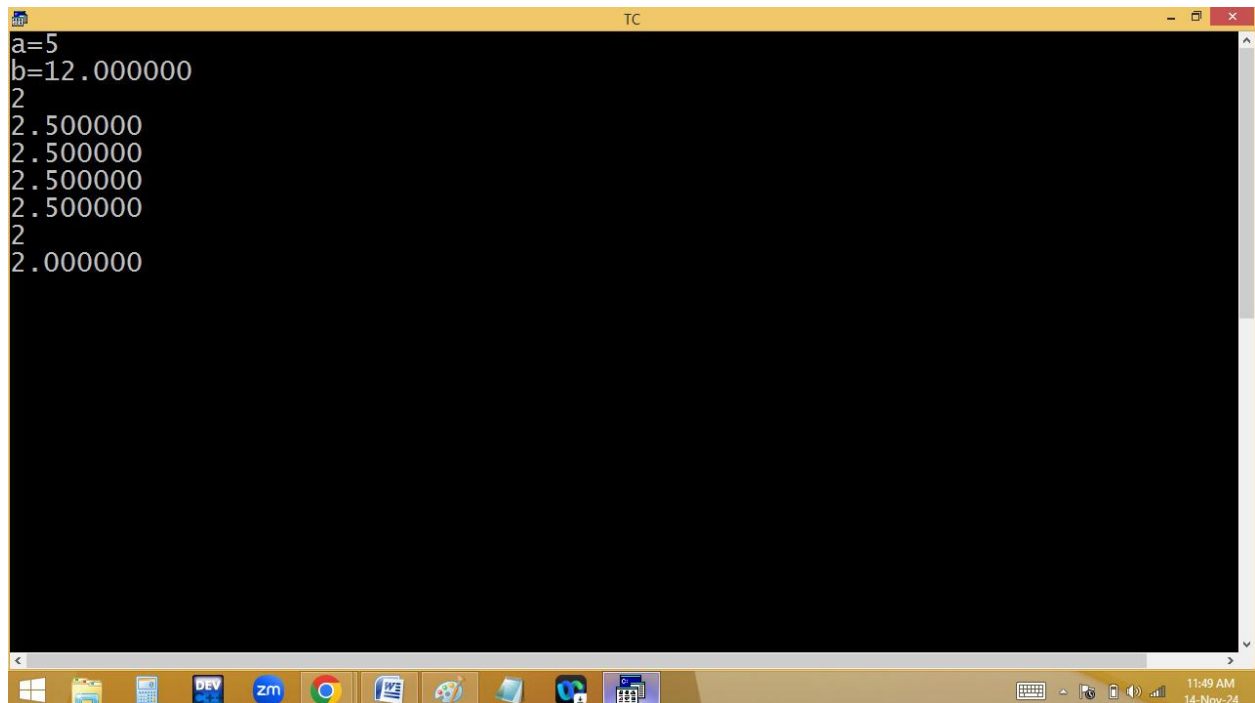
9:2

F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu
x = 2.000000_
```

(float) (5/2) = 2.000000



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 17 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=5.3; /* implicit type casting */
float b=12;
clrscr();
printf("a=%d\n",a);
printf("b=%f\n",b);
printf("%d\n", 5/2);
printf("%f\n", 5.0/2);
printf("%f\n", 5/2.0);
printf("%f\n", 5.0/2.0);
printf("%f\n", (float)5/2); /* explicit type casting */
printf("%d\n", (int)5.0/2);
printf("%f\n", (float)(5/2));
getch();
}
```



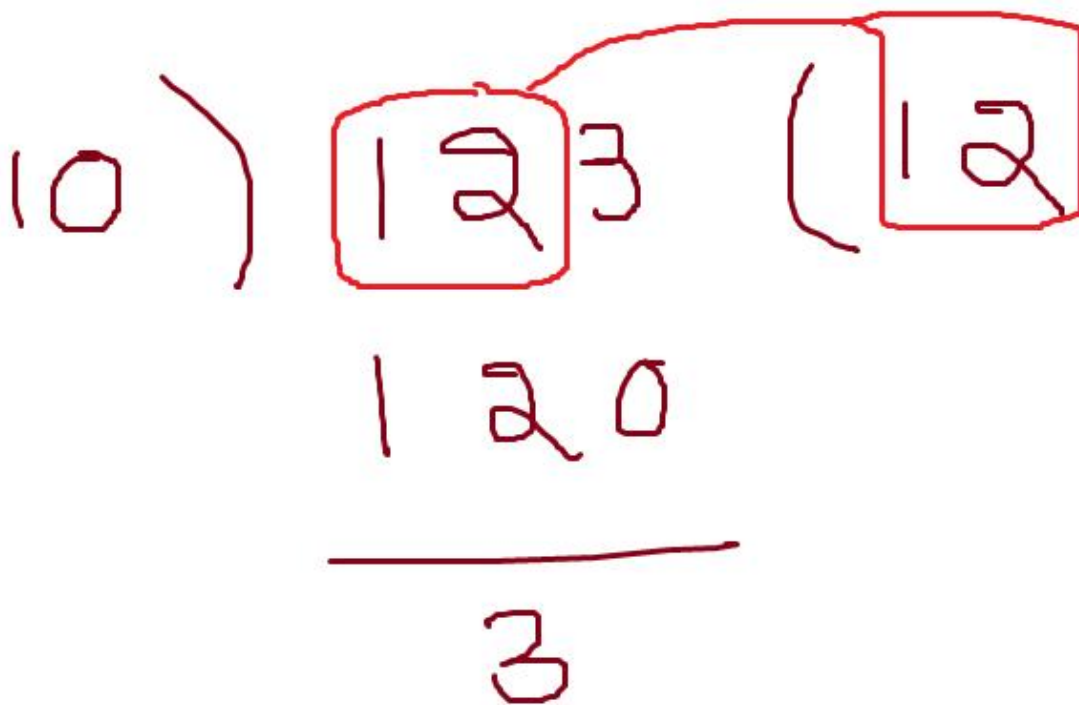
```
TC
a=5
b=12.000000
2
2.500000
2.500000
2.500000
2.500000
2
2.000000
```

12~~3~~/10=12

$$12/10=1$$

$$1/10=0$$

Note: Any no/10 removes the last digit.

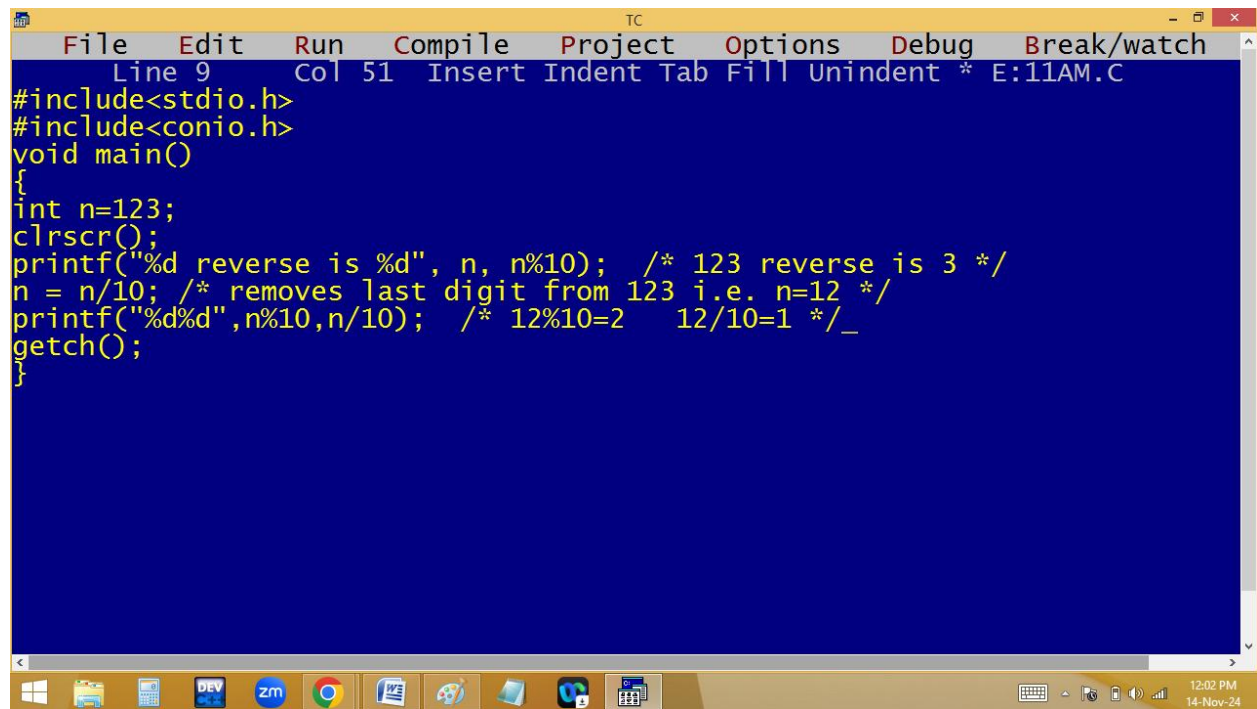


$$123\%10=3$$

Note: Any no%10 gives the last digit.

Write a C program to print a 3 digit no in reverse order without using loop.

Eg: 123 reverse is 321



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates "Line 9", "Col 51", and "Insert Indent Tab Fill Unindent * E:11AM.C". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n=123;
clrscr();
printf("%d reverse is %d", n, n%10); /* 123 reverse is 3 */
n = n/10; /* removes last digit from 123 i.e. n=12 */
printf("%d%d",n%10,n/10); /* 12%10=2 12/10=1 */
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner shows "12:02 PM" and "14-Nov-24".

```
123 reverse is 321
```

`printf ( " %d reverse is %d", n, n%10 );`

Handwritten annotations: 123 with a blue checkmark over the 3 , and $123 \% 10 = 3$ with a red arrow pointing to the $n\%10$ in the code.

$n = n/10 \implies 123/10 = 12$

Handwritten annotations: 12 and 1 with blue checkmarks, $12/10 = 1$ with a blue arrow pointing to the $n/10$ in the code, and $12 \% 10 = 2$ with a blue arrow pointing to the $n\%10$ in the code.

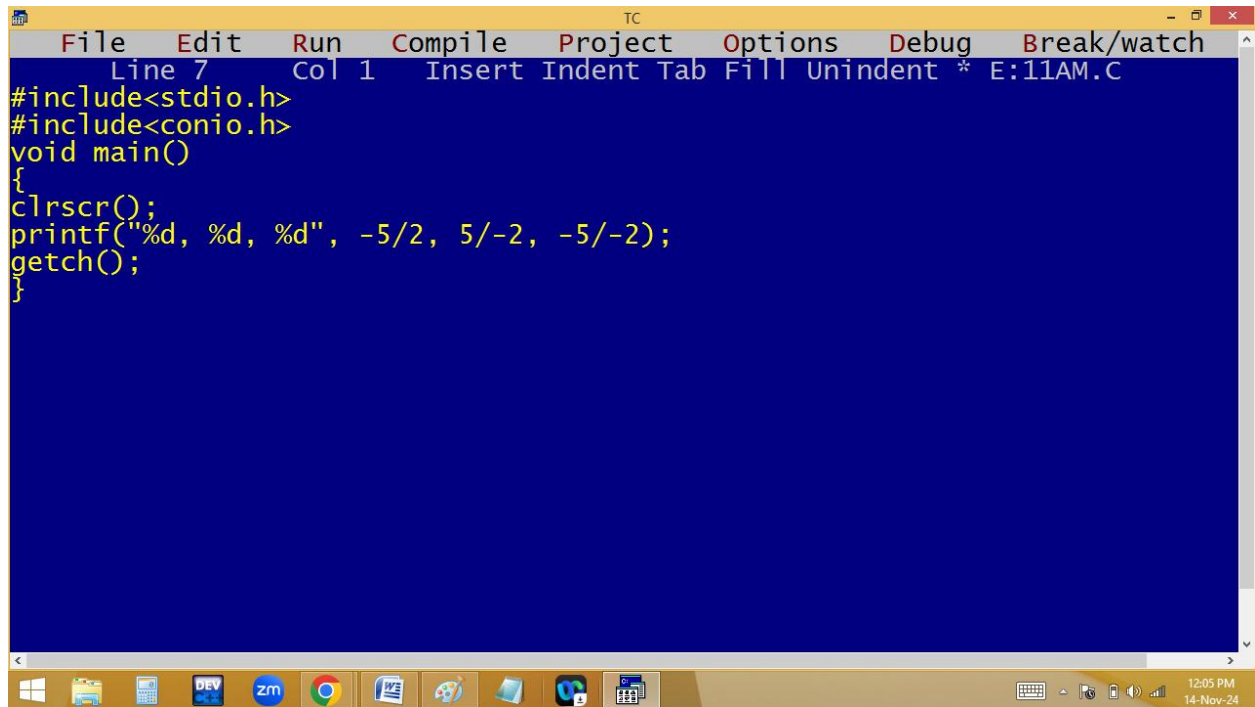
`printf("%d%d", n%10, n/10 );`

$$-5/2 = -2$$

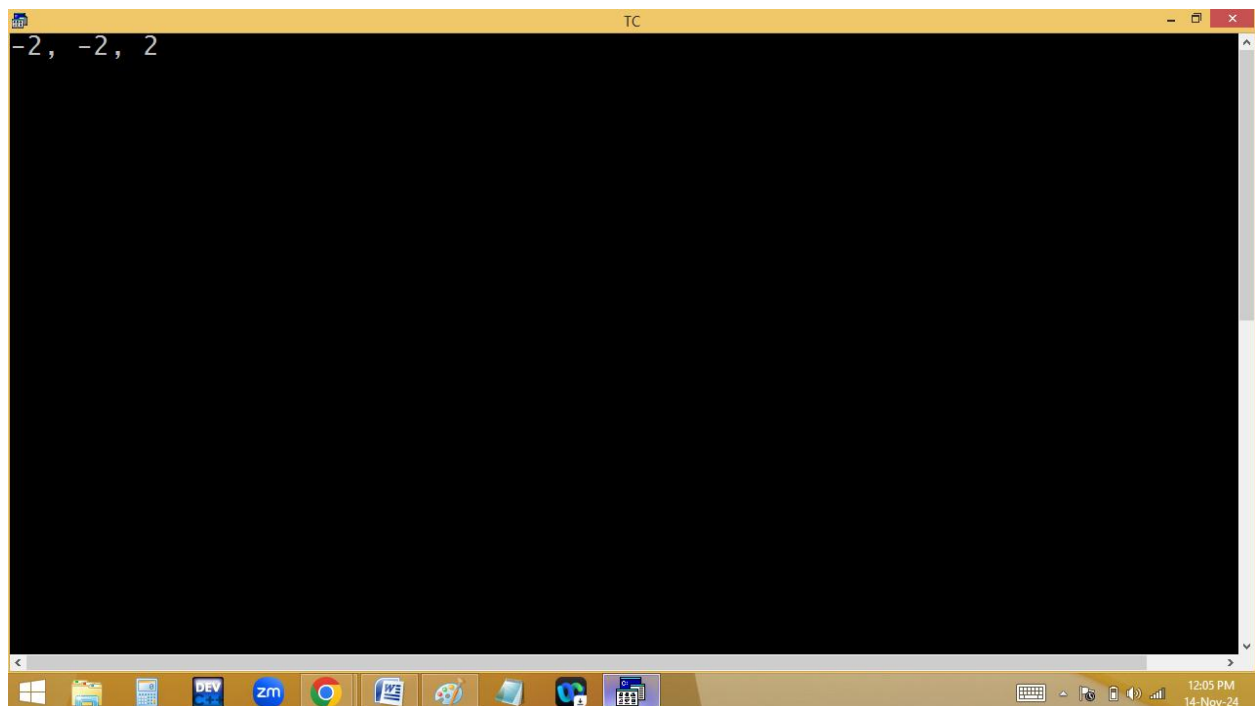
$$5/-2 = -2$$

$$-5/-2 = 2$$

Note: In division any one operand is negative then result also negative. If both are negative then result is positive.



```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d, %d, %d", -5/2, 5/-2, -5/-2);
getch();
}
```

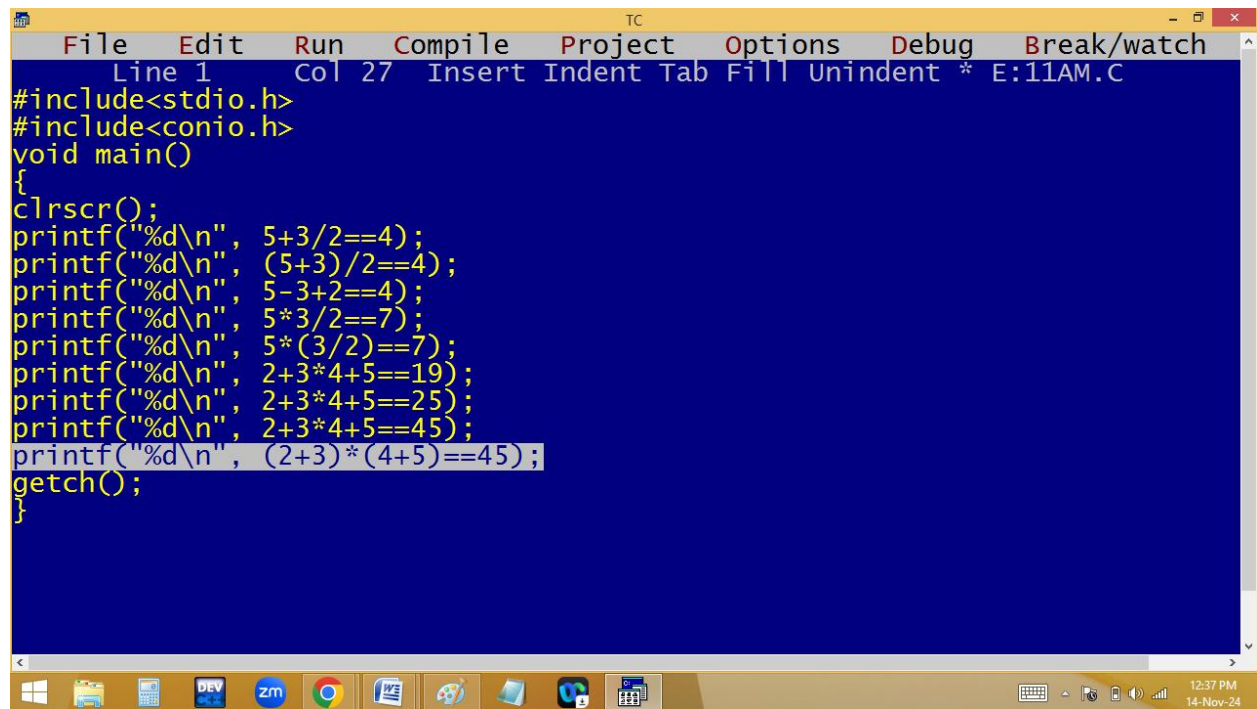


```
-2, -2, 2
```

Relational operators [== (comparison), <, >, <=, >=, !=(not equal)]: They are used to find the relation / expression / condition is true or false. If condition true always the result is 1 and false it return 0.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 31 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5==5);
printf("%d\n", 5==5.5);
printf("%d\n", 5==5.0);
printf("%d\n", 'a'==97);
printf("%d\n", 'a'>'B'); /* 97 > 66 */
printf("%d\n", 0<='0'); /* 0 <= 48 */
printf("%d\n", 'A'!=65);
printf("%d\n", 'a'>='A'+32); /* 97 >= 65+32 */
printf("a addr=%u, a addr=%u\n", "a", "a");
printf("%d\n", "a"=="a");
getch();
}
```

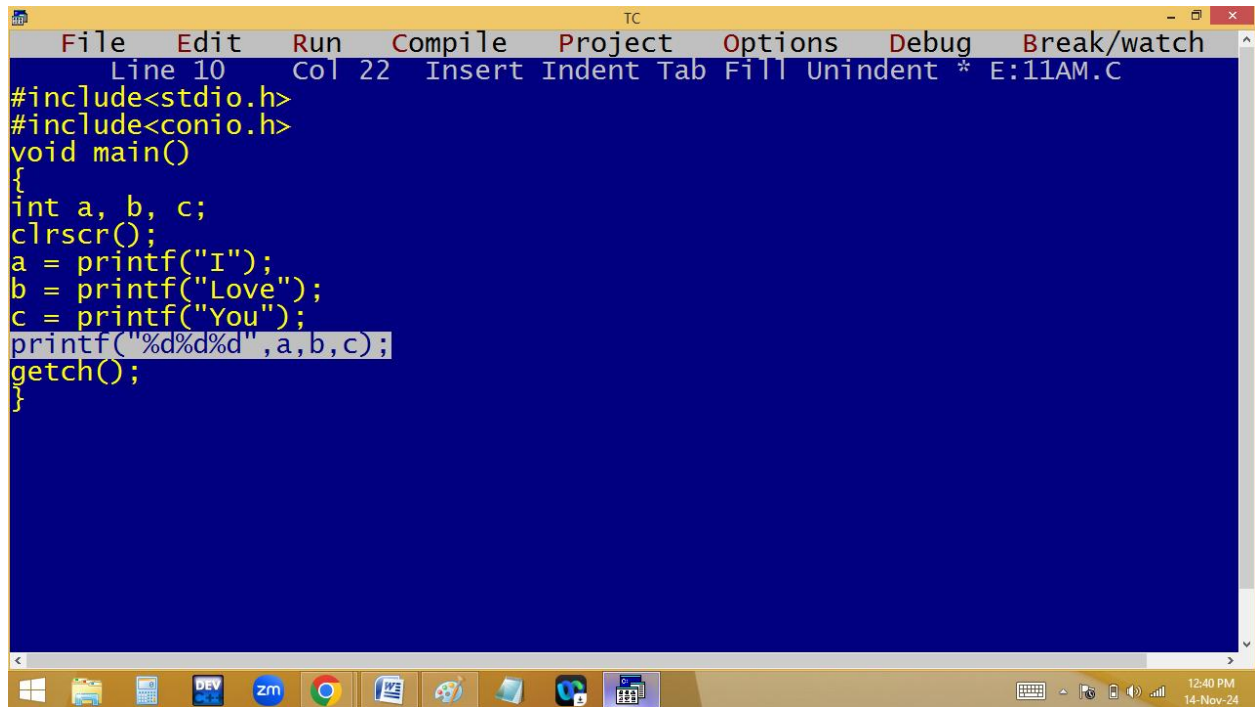
```
TC
1
0
1
1
1
1
1
0
1
a addr=458, a addr=460
0
```



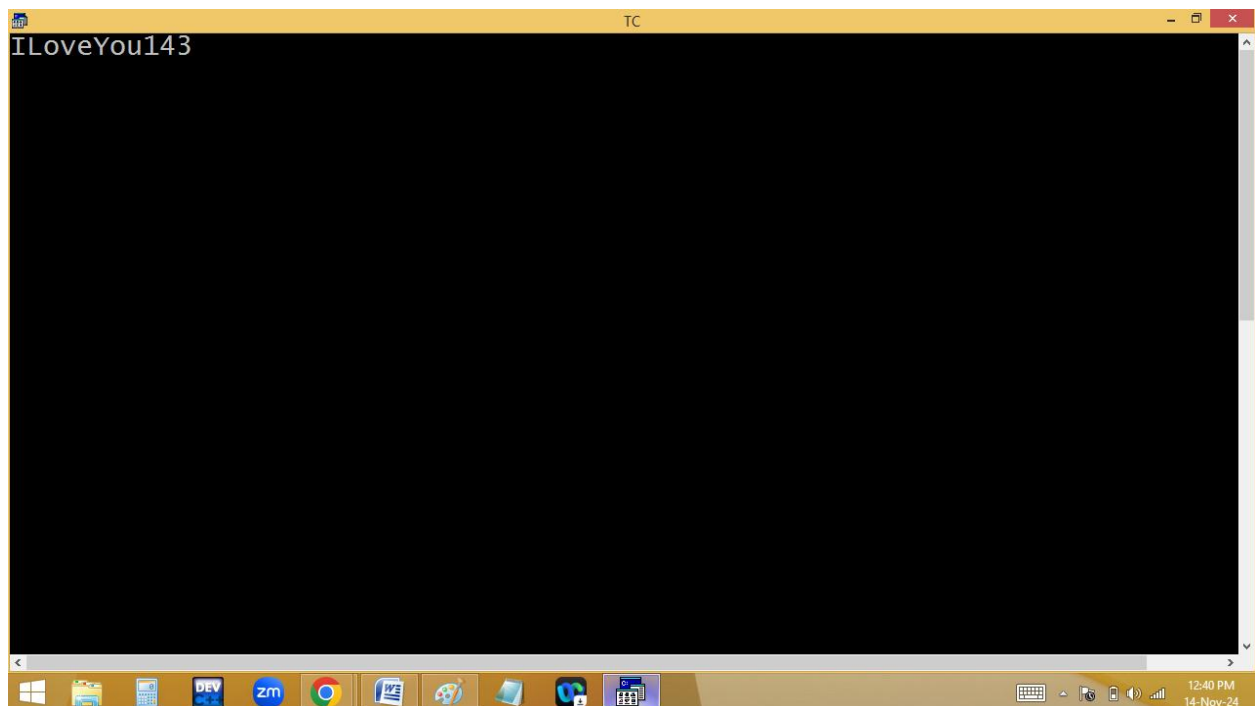
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 1", "Col 27", and "Insert Indent Tab Fill Unindent * E:11AM.C". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5+3/2==4);
printf("%d\n", (5+3)/2==4);
printf("%d\n", 5-3+2==4);
printf("%d\n", 5*3/2==7);
printf("%d\n", 5*(3/2)==7);
printf("%d\n", 2+3*4+5==19);
printf("%d\n", 2+3*4+5==25);
printf("%d\n", 2+3*4+5==45);
printf("%d\n", (2+3)*(4+5)==45);
getch();
}
```

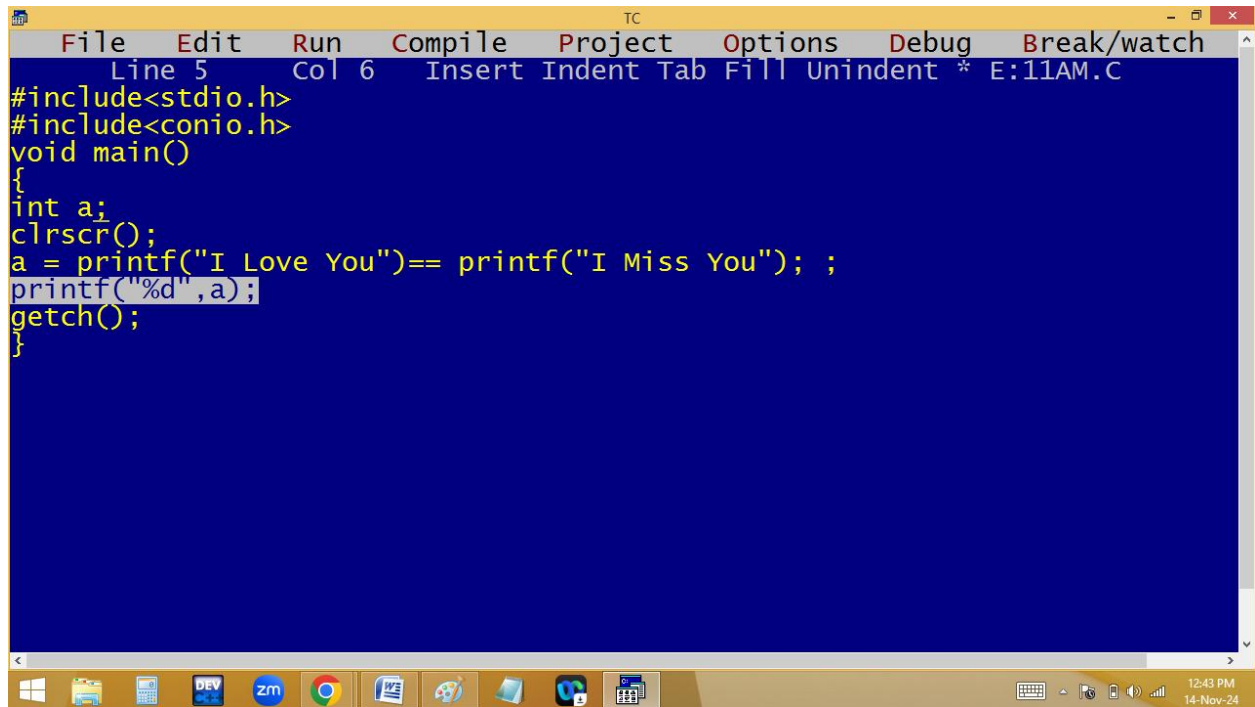
The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and other applications. The system clock in the bottom right corner shows "12:37 PM" and "14-Nov-24".



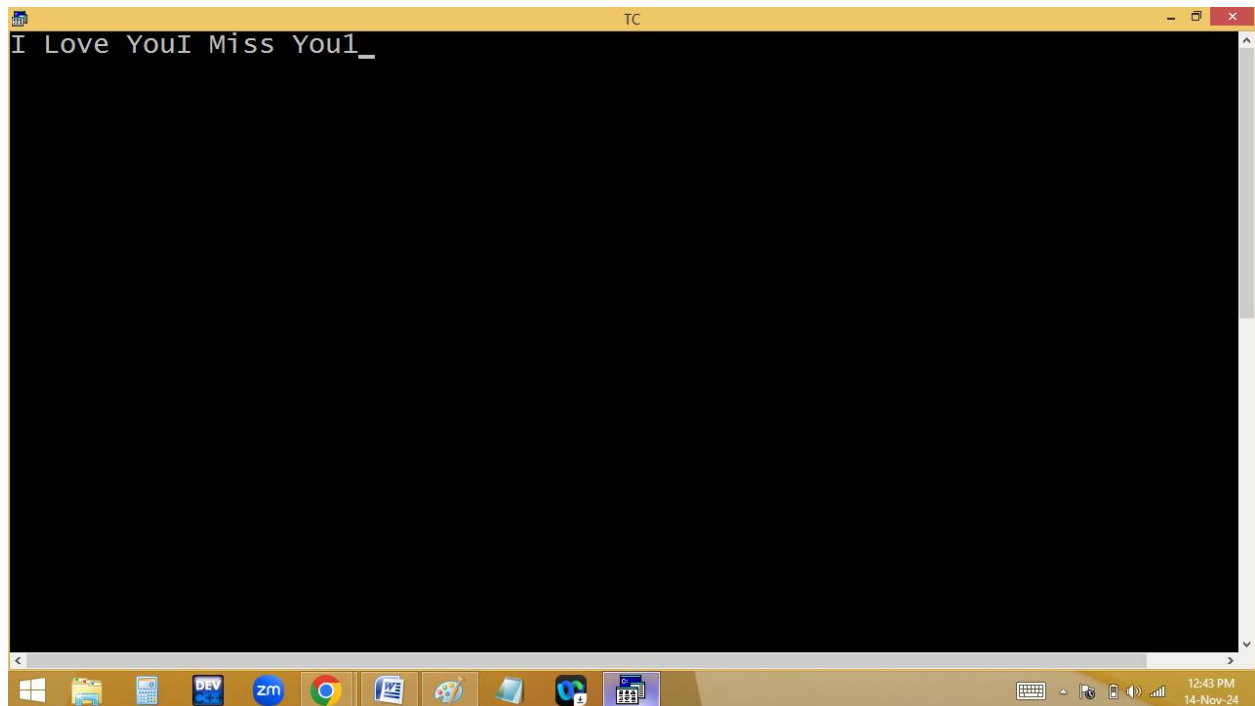
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 22 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b, c;
clrscr();
a = printf("I");
b = printf("Love");
c = printf("You");
printf("%d%d%d", a, b, c);
getch();
}
```



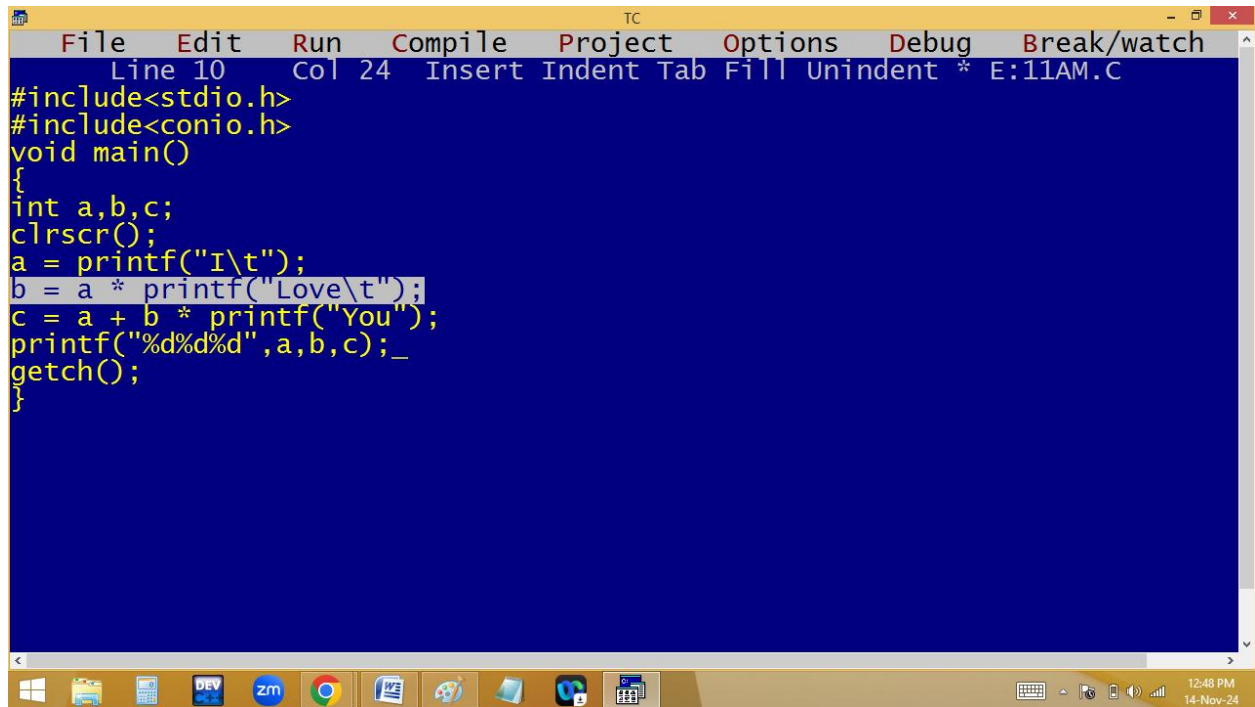
```
TC
ILoveYou143
```



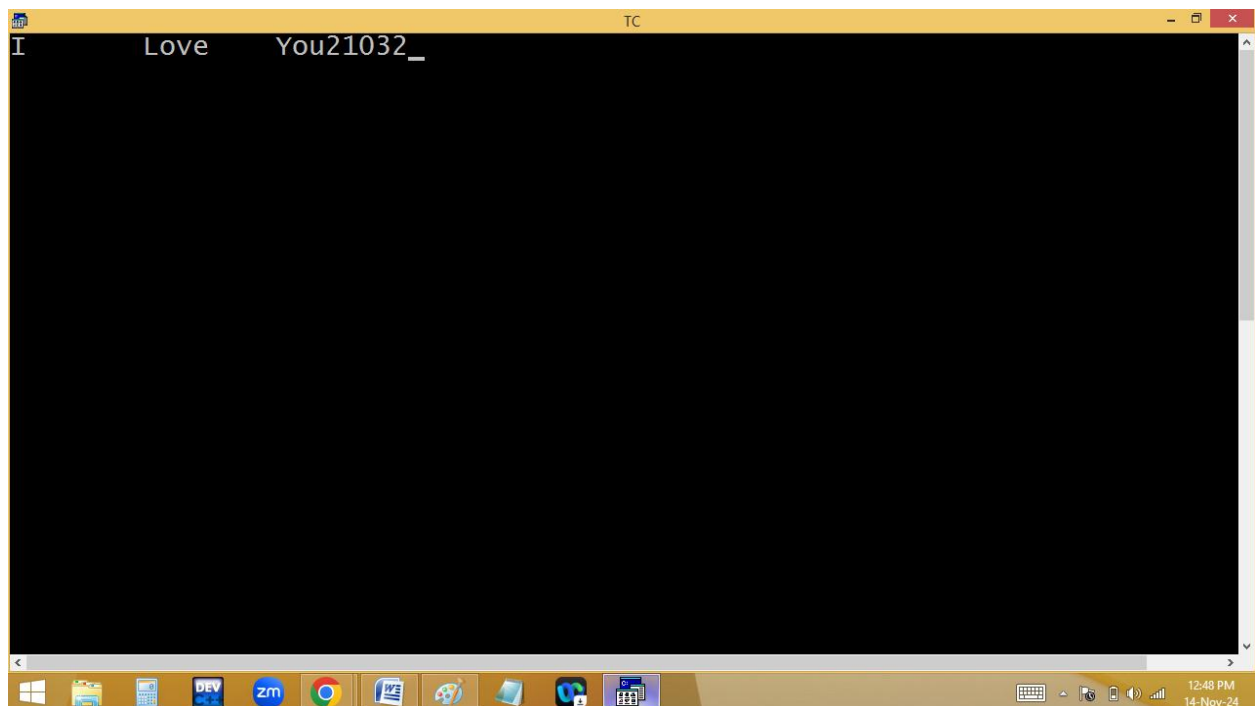
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 6 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
a = printf("I Love You")== printf("I Miss You"); ;
printf("%d",a);
getch();
}
```



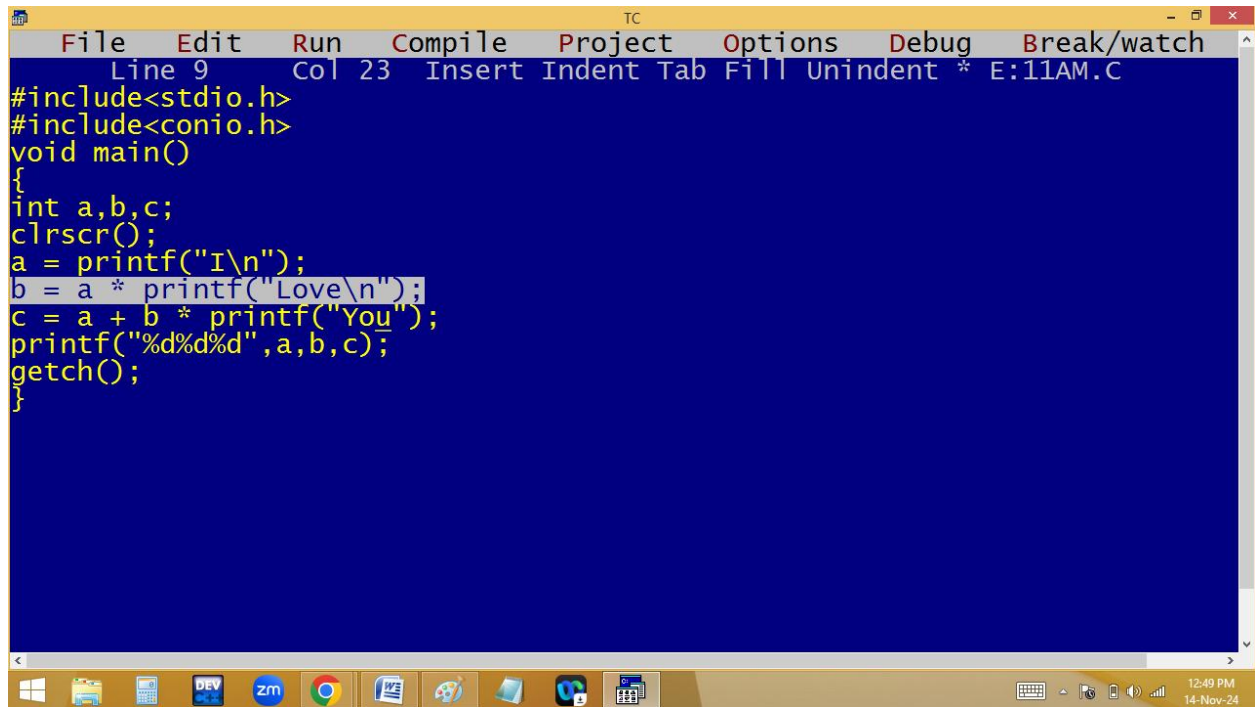
```
TC
I Love YouI Miss You1_
```



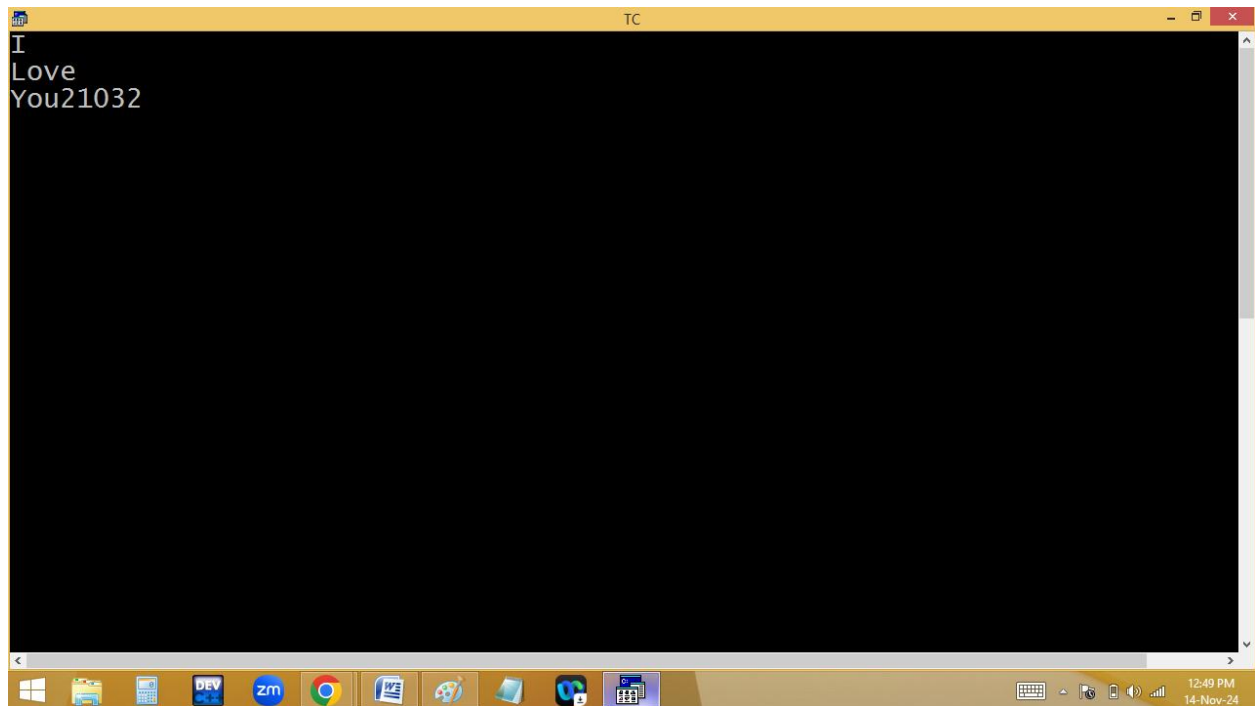
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 24 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
a = printf("I\\t");
b = a * printf("Love\\t");
c = a + b * printf("You");
printf("%d\\d\\d",a,b,c);_
getch();
}
```



```
TC
I Love You21032_
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 23 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
clrscr();
a = printf("I\\n");
b = a * printf("Love\\n");
c = a + b * printf("You");
printf("%d%d%d",a,b,c);
getch();
}
```



```
TC
I
Love
You21032
```

